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PROJECT TITLE

ML-Based Student Academic Performance Prediction System

Review-1 Project Document

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1. ABSTRACT

Student academic performance is a key indicator of educational progress. Manual review of grades, attendance and student circumstances can make it difficult to identify learners who may need support early enough. This project presents an ML-Based Student Academic Performance Prediction System that uses historical academic and selected personal/support attributes to classify a student as **Good Performance** or **At Risk**.

The prototype uses the UCI Student Performance mathematics data. The working project CSV contains 395 student records; the binary target is derived from final grade G3, with G3 >= 10 labeled Good Performance and G3 < 10 labeled At Risk. G3 is excluded from model inputs to reduce target leakage. The input set uses previous grades, study time, absences, past failures, age, family support, internet access, sex, paid classes and activities. The implemented transformation adds grade trend, weighted and average grades, support and risk-related features.

TabPFN (Tabular Prior-Data Fitted Network) is used for tabular classification. The project code describes stratified train/holdout splitting, five-fold cross-validation for configuration selection, and threshold selection focused on the At Risk class. The supplied abstract reports holdout accuracy of 89.87%, precision of 82.14%, recall of 88.46% and F1-score of 85.19%. These are reproduced as project-reported values, not independently verified in this document.

The application code includes individual prediction, probability-based risk scores, a teacher dashboard and a what-if simulator. The project abstract also describes explanations, academic trends and intervention recommendations. The system is intended as decision support for educators; its predictive output and what-if scenarios should inform human review rather than make consequential decisions automatically.

2. PROBLEM STATEMENT

Student performance monitoring often depends on manual inspection of academic records and attendance. When educators are responsible for many students, this process can delay identification of learners whose grades or engagement suggest they may need help. Support offered only after a poor final result can arrive too late to address the learning difficulties that contributed to it.

Machine learning can combine several indicators to estimate the likelihood of an at-risk outcome. However, an early-warning model is useful only when the prediction is available at a clearly defined time, avoids target leakage, and is presented in a form educators can interpret. A binary label alone does not explain the degree of risk, the attributes associated with that estimate, or how selected input changes affect the model output.

This project develops a small-data tabular prediction workflow using the UCI Student Performance mathematics records and TabPFN. It aims to present a Good Performance / At Risk classification, a probability-derived risk score, risk grouping, and what-if comparisons through a web interface. Because the implemented inputs include G1 and G2, the intended decision point is after those grade checkpoints are available; this is not a start-of-term prediction. Final grade G3 defines the target and is not supplied as a model input.

3. OBJECTIVES

The project objectives are to:

- Develop a machine-learning pipeline that predicts Good Performance versus At Risk using the UCI Student Performance dataset.
- Prepare and transform academic, attendance, behavioral and support-related attributes while excluding G3 from predictor inputs.
- Use TabPFN for classification and evaluate the project using stratified validation and a separate holdout set.
- Provide a student-level prediction and probability-based risk score that can help educators prioritize review.
- Present associated factors and academic trends in a form understandable to educators; explanations are aids to review, not causal claims.
- Offer model-based what-if scenarios for selected attributes and support timely intervention planning.
- Build an interactive Flask application with a prediction interface and a teacher-facing dashboard.

4. LITERATURE SURVEY

4.1 Overview

The project literature survey considers student performance prediction, educational data mining, early-warning systems, interpretable prediction and fairness. The supplied studies cover grade prediction, performance classification, risk identification, online/offline learning comparison and early warning. Systematic reviews by Alalawi et al. and Balaji et al. describe broad method and feature trends [1], [2].

The papers include classical methods such as logistic regression, decision trees, random forests and support vector machines; neural approaches such as attention-based BiLSTM and spiking neural networks; and interpretable rule-based models. Recent studies add feature attribution, fairness checks, threshold selection and counterfactual explanations [3]-[11]. Their results use different populations, target definitions and protocols, so published scores are not treated as a direct model ranking.

This review focuses on the connection between prediction and usable academic support. The project opportunity is to combine a data-based at-risk estimate with risk scoring, explanatory context, intervention suggestions and what-if review in one student-support workflow. The proposed contribution is integration and application; it does not claim that TabPFN is universally superior.

4.2 Literature Survey					
S.No.	**Research Paper**	**Model / Approach**	**Dataset / Data Used + Link**	**Key Outcome**	**Limitation**
1	Alalawi, Athauda & Chiong (2023). *Contextualizing the current state of research on the use of machine learning for student performance prediction: A systematic literature review.*	Systematic literature review using defined search and inclusion criteria to examine student performance and dropout prediction.	162 papers published from Jan. 2010 to Oct. 2022; review study, no primary dataset. https://doi.org/10.1002/eng2.12699	Classifies the studies into performance/at-risk prediction and dropout/retention prediction; reports 129 and 33 studies in those groups.	Maps the research field but does not implement an individual student-facing prediction and intervention application.
2	Kaur (2025). *Prediction of Student Academic Performance in Higher Education Using Machine Learning Techniques.*	Decision Tree, Random Forest, SVM and Logistic Regression applied to historical academic records.	Historical academic and demographic data; the supplied paper does not clearly identify a reusable dataset. Source paper	The paper states an expected 70-85% accuracy range. It is an expectation, not a measured result.	Dataset details, validation protocol and empirical evidence are not clearly documented in the supplied copy.
3	Yousafzai et al. (2021). *Student-Performulator: Student Academic Performance Using Hybrid Deep Neural Network.*	Attention-based BiLSTM with feature selection to predict student grade categories.	Historical student academic records used by the study. https://doi.org/10.3390/su13179775	The paper reports 90.16% accuracy for its grade-prediction task.	Focuses on grade prediction; a student-level risk score and educator intervention workflow are not described.

4	Liu et al. (2022). *A Predictive Model for Student Achievement Using Spiking Neural Networks Based on Educational Data.*	Evolutionary spiking neural network; membrane algorithm tunes model hyperparameters.	UCI Student Performance mathematics data (395 records) and xAPI-Edu-Data (480 records). https://doi.org/10.3390/app12083841	Evaluates prediction on two educational datasets and reports comparisons with other methods. Results are dataset/task-specific.	Prediction modeling is central; explanations and support actions are not integrated into an educator-facing workflow.
5	Holicza & Kiss (2023). *Predicting and Comparing Students Online and Offline Academic Performance Using Machine Learning Algorithms.*	SVM, Decision Tree, Random Forest and KNN, applied to online and offline student records.	Two datasets describing online-learning and offline-learning contexts. https://doi.org/10.3390/bs13040289	Compares accuracy and F1 and discusses sleep, study time and screen-time habits in relation to performance.	Compares data contexts and predictors; no integrated risk score and intervention application is described.
6	Zhao et al. (2023). *Quantitative Analysis and Prediction of Academic Performance of Students Using Machine Learning.*	Multiple machine-learning classifiers with feature and influencing-factor analysis.	Collected educational data described in the study. https://doi.org/10.3390/su151612531	Analyzes prediction of academic performance and the relationship between student factors and outcomes.	Factor analysis does not itself provide a student-level explanation, intervention recommendation or monitoring workflow.
7	Liu et al. (2024). *A New Student Performance Prediction Method Based on Belief Rule Base with Automated Construction.*	Auto-BRB with evidential reasoning, P-CMA-ES optimization and AIC model selection.	Student-performance case studies, including secondary education and college English. https://doi.org/10.3390/math12152418	Authors report an accuracy advantage over traditional models while retaining interpretability.	An interpretable model is proposed, but risk scores, educator interventions and an interactive what-if flow are not described.

8	Kesgin et al. (2025). *Beyond Performance: Explaining and Ensuring Fairness in Student Academic Performance Prediction with Machine Learning.*	Logistic Regression, Random Forest and XGBoost with resampling, SHAP and fairness assessment.	UCI Student Performance records; subgroup evaluation is included. https://doi.org/10.3390/app15158409	Reports XGBoost accuracy 0.789 and F1 0.803 under the paper's evaluation protocol.	Strong explanation/fairness scope; does not describe an integrated intervention and what-if application.
9	El Jihaoui, Abra & Mansouri (2026). *Towards a Fair and Transparent Early Warning System for Identifying Students at Risk of Academic Failure.*	Calibrated models, recall-constrained threshold selection, SAGE/ALE, DiCE and intersectional fairness audit.	451,852 middle-school records; first-semester data predict year-end Pass/Fail for one regional cohort. https://doi.org/10.3389/feduc.2026.1878686	Reports MLP recall 0.867, precision 0.751, AUC-PR 0.888 and MCC 0.730 at its selected operating point.	An integrated early-warning approach, but its population and dataset differ from this project.
10	Vecchi (2026). *Investigating the Efficacy and Interpretability of ML Classifiers for Student Performance Prediction in the Small-Data Regime.*	Comparative supervised classifiers, feature selection and interpretable-model analysis.	Two-high-school dataset; binary performance classification in a small-data setting. https://doi.org/10.3390/educsci16010149	Reports robust performance for several classifiers and examines important predictors; no single abstract-level score is quoted here.	Focuses on model and feature behavior; calls for richer longitudinal and psychological data.

5. RESEARCH GAP IDENTIFICATION

5.1 Identified Research Gaps

The literature survey shows that student performance prediction is studied through different data sources, targets and modeling approaches. However, model outputs do not automatically become timely and understandable academic support. The following research gaps were identified within the reviewed scope:

S.No.	Research Gap
1.	Limited connection between prediction and support: Many studies focus on grades, classification or risk identification, while fewer describe how an individual prediction is connected to an educator review and timely academic support.
2.	Different datasets and target definitions: Studies use distinct populations, grade categories, predictors and validation procedures. Their performance values cannot be treated as a common ranking or assumed to transfer to the UCI mathematics cohort.
3.	Limited integrated risk presentation: A predicted class alone may not communicate the risk probability, the associated student attributes or the basis for prioritizing a case.
4.	Limited what-if use in the reviewed set: Most included studies evaluate models or factors; few present a student-level interactive scenario interface alongside prediction.
5.	Need for careful prediction timing: The project inputs include G1 and G2. Therefore, the model is applicable after those checkpoint grades are available and should not be described as a start-of-term predictor.
6.	Limited external validity: The project working file contains 395 mathematics records. Performance, fairness and calibration must be examined before applying the model to other schools or cohorts.

5.2 Link Between Research Gaps and Proposed Work

The proposed system addresses these project-level gaps by connecting a TabPFN prediction with a probability-derived risk score, a teacher dashboard and selected what-if scenarios. The model uses UCI Student Performance mathematics records and a binary target derived from final grade G3. G3 is excluded from model inputs.

At the intended checkpoint, the system accepts available academic and student attributes, returns Good Performance or At Risk, and presents a risk band for educator review. The what-if interface compares the current prediction with selected changes to G2, absences and study time. These outputs support review and discussion; they do not establish that a simulated change causes a better academic result.

The contribution is this integration in a small-data prototype. Its value must be assessed through class-wise evaluation, model calibration, subgroup checks, external validation and educator feedback. The comparison therefore motivates the proposed workflow without claiming that TabPFN is more accurate than models evaluated on unrelated datasets.

6. INNOVATION

The proposed system introduces an integrated approach to academic risk prediction and educator-facing review. The main innovation is the combination of tabular prediction, risk scoring, selected explanations and scenario analysis within a single student-support workflow.

The proposed innovations are:

- **TabPFN-Based Tabular Prediction**
  The project applies TabPFN to the selected UCI mathematics data and derives Good Performance / At Risk classes from the final grade.
- **Leakage-Aware Target Design**
  G3 determines the target but is excluded from model inputs, reducing direct target leakage.
- **Academic and Support Features**
  The pipeline combines prior grades, study time, absences, failures, age, family support, internet access, paid classes and activities.
- **Probability-Based Risk Score**
  The system presents an individual risk probability and groups records into low, moderate and high bands for review.
- **Interactive What-If Scenarios**
  The application compares selected changes to G2, absences and study time; these are model scenarios, not causal intervention effects.
- **Unified Educator Workflow**
  The Flask application connects a student prediction, risk view, teacher dashboard and support-oriented interpretation.

7. CREATIVITY

The project introduces a learner-support view where educators can review a risk prediction together with relevant academic information. The creative aspect lies in combining model outputs with interactive and practical monitoring activities.

The main creative elements are:

- **Individual Risk Card**
  The interface can show the predicted class, risk probability and risk band together for a student record.
- **Academic Trend Context**
  G1 and G2 are available beside the prediction so educators can interpret the score in relation to recent grades.
- **Teacher Dashboard**
  Records are organized by risk level and sorted to help educators prioritize which cases to review.
- **Interactive What-If Analysis**
  The user can compare a current prediction with bounded changes to selected inputs and review how the model output changes.
- **Support-Oriented Recommendations**
  Risk signals can prompt educator consideration of tutoring, attendance follow-up or other support; the final action remains a human decision.
- **Single Browser Workflow**
  A Flask interface provides individual prediction and a dashboard without requiring educators to run model scripts.

8. NOVELTY

The novelty of the proposed work lies in integrating a tabular prediction model with a risk-oriented academic review workflow. TabPFN, machine-learning prediction and dashboards are established techniques; the project contribution is their application and integration for the selected student-performance use case.

The novel aspects of the proposed work are:

- **Prediction with Risk Context**
  Combines a Good Performance / At Risk label with a probability-derived risk score and risk band.
- **Checkpoint-Based Use**
  Defines the prediction time after G1 and G2 are available instead of presenting the model as a beginning-of-term forecast.
- **Leakage-Aware Target Handling**
  Uses G3 only to define the outcome and excludes it from model predictors.
- **Scenario Review**
  Places model-based what-if results alongside the current estimate to support educator discussion.
- **Integrated Student-Support View**
  Connects prediction, dashboard review, explanatory context and support suggestions in the same workflow.
- **Transparent Contribution Boundary**
  Frames the contribution as system integration; a claim of improved accuracy requires common datasets, baselines and validation.

9. FEASIBILITY

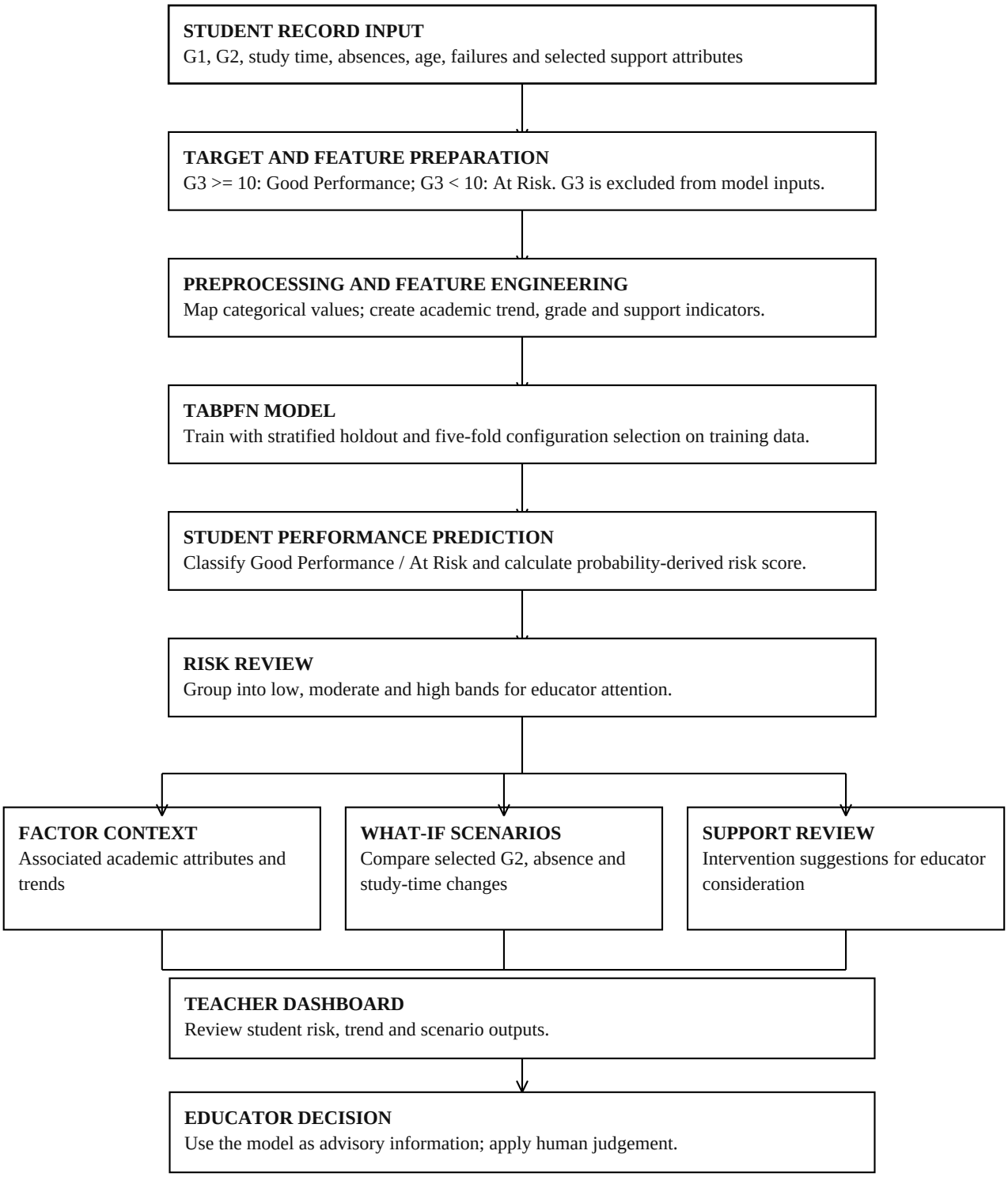
The proposed system is feasible to develop using currently available software technologies, a public educational dataset and an existing Python project archive. The project can be implemented as a web-based application with separate data, model and interface components.

The feasibility of the project is considered in the following aspects:

- **Technical**
  The repository includes Python data transformation, TabPFN training/prediction and Flask application code. pandas and scikit-learn support data handling and evaluation.
- **Data**
  The working UCI mathematics file contains 395 records. A selected set of academic, attendance, behavioral and support-related attributes is available.
- **Computational**
  The dataset is small enough for prototype experiments on available CPU hardware. Model version and runtime requirements should be documented.
- **Implementation**
  The archive includes individual prediction, a teacher dashboard and four selected what-if cases. Explanations and recommendations require end-to-end review.
- **Economic**
  The prototype can use open-source Python libraries and a public dataset, with no requirement for a proprietary data source.
- **Time**
  The work can proceed through literature review, data checks, model reconciliation, application refinement, evaluation and documentation.

10. PROPOSED FRAMEWORK

The proposed system provides a checkpoint-based workflow for classifying student performance and supporting educator review. The final grade is used to define the training target and is kept out of the predictor inputs.



10.1 FRAMEWORK DESCRIPTION

The project working CSV contains 395 records and uses study time, absences, G1, G2, age, prior failures, family support, internet access, sex, paid classes and activities.

1. Define the target. Derive Good Performance for G3 >= 10 and At Risk for G3 < 10. Drop G3 from X.

2. Transform features. Convert categorical values to numeric form. The code builds 19 model columns from 11 selected source attributes plus eight engineered features.

3. Prepare evaluation data. The top-level training script uses a stratified 80/20 train/holdout split with random state 42. Five-fold stratified CV selects a configuration and threshold on the training split.

4. Fit and save TabPFN. Train the selected configuration, save the estimator and metadata, then apply the saved threshold.

5. Produce a student result. Return class and risk probability. Dashboard bands are Low below 25%, Moderate from 25% to below 60%, and High at 60% or above.

6. Review scenarios. Evaluate four input changes: G2 +2, absences -5, study time +1, and G2 +2 with absences -5. These are model what-if outputs.

The archive contains two training entry points: a component trainer fixes TabPFN n_estimators at 8, while train.py searches 4, 8 and 16. They should be reconciled into one documented training command before final results are treated as canonical.

11. PROJECT PLAN

The work proceeds from source and data verification through modeling, application integration and evaluation.

11.1 Project Execution Phases

Phase	Work to be Completed
Phase 1	Confirm objectives, dataset provenance, target and prediction checkpoint.
Phase 2	Audit UCI source and working CSV; verify rows, labels, missing values and input fields.
Phase 3	Implement preprocessing and feature engineering; exclude G3 and student_id from model inputs.
Phase 4	Train TabPFN with fixed split and stratified cross-validation for configuration selection.
Phase 5	Evaluate holdout class-wise metrics, confusion matrix and probability calibration.
Phase 6	Finalize Flask prediction, risk-score display, teacher dashboard and what-if simulator.
Phase 7	Review explanation/recommendation behavior, subgroup performance, privacy and usability.
Phase 8	Reconcile abstract/code metrics, document limitations and prepare final presentation.

11.2 Complete Workflow of Execution

Literature Survey

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Research Gap Identification

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Dataset and Project-File Verification

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Target Definition and Leakage Check

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Data Preprocessing

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Feature Engineering

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Stratified Train/Holdout Split

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TabPFN Configuration Selection with 5-Fold CV

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Final Model Training and Evaluation

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Flask Prediction and Risk-Score Display

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Dashboard and What-If Review

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Educator Feedback, Final Testing and Documentation

11.3 Expected Evaluation

The developed system should be evaluated based on:

- At Risk precision, recall and F1-score, along with accuracy and the confusion matrix
- Performance at the selected risk threshold and the number of missed at-risk students
- Probability calibration and the behavior of low, moderate and high risk bands
- Consistency of results across the available student subgroups
- Clarity and usefulness of displayed academic factors and trends
- Relevance of intervention suggestions and usefulness of what-if review to educators
- Generalization to a separate cohort before any broader deployment

The supplied project abstract reports holdout results of 89.87% accuracy, 82.14% precision, 88.46% recall and 85.19% F1. These values are reproduced as abstract-reported results. The archived model_results.csv separately records training-set five-fold configuration-selection metrics: for the 16-estimator configuration, 90.51% CV accuracy, 86.73% At Risk F1 and 94.23% At Risk recall at threshold 0.33. These two summaries describe different evaluation stages and must not be combined.

The archive also contains a component trainer fixed at 8 estimators and a top-level training script that searches 4, 8 and 16. The project should reconcile these paths, rerun one documented pipeline, and report its split, threshold, class averaging and random seed before making a final performance claim.

12. CONCLUSION

This project presents an ML-based approach for identifying students who may be at risk of low academic performance using a small, public student dataset. It defines a binary target from G3, excludes G3 from predictors, engineers academic and support-related features, and uses TabPFN for classification. The Flask prototype includes individual prediction, risk scoring, a teacher dashboard and what-if scenarios.

The contribution is connecting a risk estimate with a workflow for prioritizing attention and reviewing selected scenarios. The model does not establish that an intervention will improve outcomes, and a probability score is not a causal explanation.

The supplied abstract reports holdout metrics, while the archived CSV records a separate set of cross-validation configuration scores. The training paths must be reconciled and a canonical evaluation rerun before making a final performance claim. Future work should include common baselines, class-wise reporting, calibration, subgroup fairness checks, external validation, privacy protections and educator evaluation. With those safeguards, the system can serve as decision support for timely academic review rather than an automated decision-maker.

13. REFERENCES

[1] Alalawi, K., Athauda, R., & Chiong, R. (2023). Contextualizing the current state of research on the use of machine learning for student performance prediction: A systematic literature review. *Engineering Reports, 5*(12), e12699. https://doi.org/10.1002/eng2.12699

[2] Balaji, P., Alelyani, S., Qahmash, A., & Mohana, M. (2021). Contributions of machine learning models towards student academic performance prediction: A systematic review. *Applied Sciences, 11*(21), 10007. https://doi.org/10.3390/app112110007

[3] Kaur, B. (2025). Prediction of student academic performance in higher education using machine learning techniques. *International Journal for Multidisciplinary Research, 7*(6). Supplied paper labels 70-85% as an expected range.

[4] Yousafzai, B. K., Khan, S. A., Rahman, T., Khan, I., Ullah, I., Ur Rehman, A., Baz, M., Hamam, H., & Cheikhrouhou, O. (2021). Student-Performulator: Student academic performance using hybrid deep neural network. *Sustainability, 13*(17), 9775. https://doi.org/10.3390/su13179775

[5] Liu, C., Wang, H., Du, Y., & Yuan, Z. (2022). A predictive model for student achievement using spiking neural networks based on educational data. *Applied Sciences, 12*(8), 3841. https://doi.org/10.3390/app12083841

[6] Holicza, B., & Kiss, A. (2023). Predicting and comparing students online and offline academic performance using machine learning algorithms. *Behavioral Sciences, 13*(4), 289. https://doi.org/10.3390/bs13040289

[7] Zhao, L., Ren, J., Zhang, L., & Zhao, H. (2023). Quantitative analysis and prediction of academic performance of students using machine learning. *Sustainability, 15*(16), 12531. https://doi.org/10.3390/su151612531

[8] Liu, M., He, W., Zhou, G., & Zhu, H. (2024). A new student performance prediction method based on belief rule base with automated construction. *Mathematics, 12*(15), 2418. https://doi.org/10.3390/math12152418

[9] Kesgin, K., Kiraz, S., Kosunalp, S., & Stoycheva, B. (2025). Beyond performance: Explaining and ensuring fairness in student academic performance prediction with machine learning. *Applied Sciences, 15*(15), 8409. https://doi.org/10.3390/app15158409

[10] El Jihaoui, M., Abra, O. E. K., & Mansouri, K. (2026). Towards a fair and transparent early warning system for identifying students at risk of academic failure. *Frontiers in Education, 11*, 1878686. https://doi.org/10.3389/feduc.2026.1878686

[11] Vecchi, E. (2026). Investigating the efficacy and interpretability of ML classifiers for student performance prediction in the small-data regime. *Education Sciences, 16*(1), 149. https://doi.org/10.3390/educsci16010149

[12] Cortez, P., & Silva, A. M. G. (2008). Using data mining to predict secondary school student performance. In *Proceedings of the 5th Future Business Technology Conference*. UCI Student Performance dataset, repository record 320.

[13] Hollmann, N., Muller, S., Eggensperger, K., & Hutter, F. (2023). TabPFN: A transformer that solves small tabular classification problems in a second. *International Conference on Learning Representations*. arXiv:2207.01848.

[14] Project source materials supplied by the team: Project_Abstract.docx; Project_PPT_Review1.pptx; ALT PROJECT source archive, including transformation, training, prediction, Flask application and model_results.csv files.